

COSC420: Assignment 1 Report

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1 Introduction

This report presents two deep learning models trained on the Oxford Flower Datasets. The first is a CNN classifier designed for flower classification. The second is a Latent Stable Diffusion Model to generate flower-like images from noise, consisting of a VAE and a UNet Denoiser. This report details the model architectures, training and validation, and also evaluates their test result performance.

2 CNN Classifier

This CNN classifier has two modes compatible with 10-class coarse grained and 102-class fine grained datasets, with a total of 451,530 and 498,726 learnable parameters respectively. Table 1 shows the model architecture in 10-class mode (excluding pooling and dropout layers because there's no learnable parameters), listing the layer types, output shapes, and the number of learnable parameters (namely the weights and bias).

2.1 CNN Classifier Architecture

This model consists of 17 layers (Conv2d x 9, BatchNorm x 6, Linear x 2). Batch normalization is applied to stabilize training, learning rate based on 0.001 is adaptively reduced on plateau of loss decreasing to gradually reach a local or global minimum without jumping around too fast, and dropout rate is flexibly set to reduce overfitting (0.2 and 0.3 after the first two convolutional blocks, and 0.5 after the linear layer). In addition, to address class imbalance in the training set, a weighted random sampler is used based on different flower class frequencies; while the validation and test sets are unchanged original data.

Table 1: CNN Architecture (Batch Size=16, Input Size=96×96)

Layer Type	Output Shape	Parameters
Conv2d	[16, 32, 96, 96]	896
BatchNorm2d	[16, 32, 96, 96]	64
Conv2d (Depthwise)	[16, 32, 96, 96]	320
Conv2d (Pointwise)	[16, 64, 96, 96]	2,112
BatchNorm2d	[16, 64, 96, 96]	128
Conv2d (Depthwise)	[16, 64, 48, 48]	640
Conv2d (Pointwise)	[16, 128, 48, 48]	8,320
BatchNorm2d	[16, 128, 48, 48]	256
Conv2d (Depthwise)	[16, 128, 48, 48]	1,280
Conv2d (Pointwise)	[16, 256, 48, 48]	33,024
BatchNorm2d	[16, 256, 48, 48]	512
Conv2d (Depthwise)	[16, 256, 24, 24]	2,560
Conv2d (Pointwise)	[16, 512, 24, 24]	131,584
BatchNorm2d	[16, 512, 24, 24]	1,024
Linear	[16, 512]	262,656
BatchNorm1d	[16, 512]	1,024
Linear	[16, 10]	5,130

2.2 Coarse Grained: Training, Validation, Test

The CNN classifier has been trained on coarse grained datasets for 100 epochs. At the early epochs the training and validation losses steadily decreased; and during the final 20 epochs (81–100), training loss oscillated between 0.23–0.15 and converged to 0.08, validation loss fluctuated between 1.4–0.72 while maintaining downward trend. Regarding the accuracy, the validation reached a peak of around 83% during the latest epochs, with a gap of 14% from training which indicates moderate overfitting.

For the test results, Figure 1 shows the CNN classifier achieved 85.13% accuracy on coarse grained flower classification. The best performance is observed for Composite Flowers (precision 97.10%, recall 95.71%, F1 score 96.40%) , followed by Water Flowers (precision 100%, recall 90%, F1 score 94.74%). And the lowest accuracy is found for Bell-shaped Flowers with 7 misclassifications out of 20 samples.

2.3 Fine Grained: Training, Validation, Test

The CNN classifier has been trained on fine grained datasets for 100 epochs. In the final 20 epochs (81–100), training loss steadily decreased from 0.28 to 0.11, and accuracy improved from 91.46% to 96.52%. In contrast, validation loss fluctuated between 1.19–0.72, and accuracy fluctuated between 78% to 85%, while the overall trends indicate a decreasing validation loss and a peak validation accuracy of around 85% at the end. Meanwhile, the 12% accuracy gap between training and validation shows moderate overfitting.

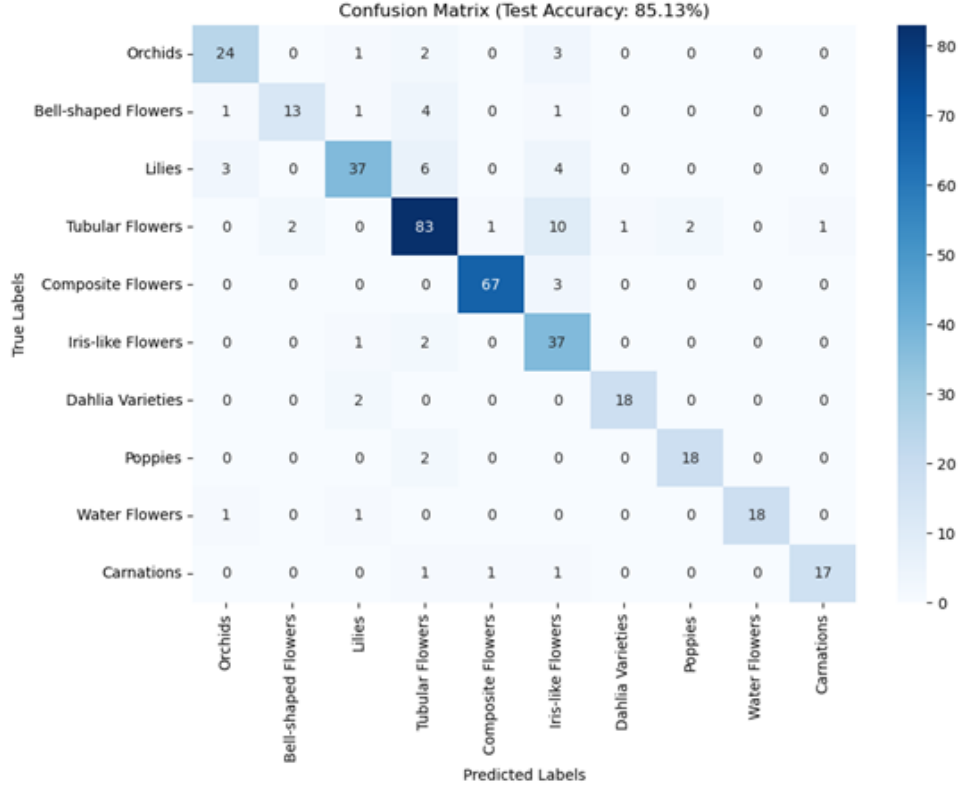


Figure 1: Coarse Grained Test Confusion Matrix

Due to the complexity of the 102-class confusion matrix, classification performance is summarized in Table 2. The macro and weighted averages of precision, recall, and F1-score range from 0.83 to 0.85, indicating consistent overall performance. However, classes like Canterbury Bells, with a low recall of 0.3, highlight opportunities for improving class-specific sensitivity.

Table 2: Fine Grained Test Classification Performance

Class	Precision	Recall	F1-Score	Support
Pink Primrose	0.8750	0.7000	0.7778	10
Hard-Leaved Pocket Orchid	0.8333	1.0000	0.9091	10
Canterbury Bells	1.0000	0.3000	0.4615	10
Sweet Pea	0.6667	0.6000	0.6316	10
English Marigold	1.0000	0.9000	0.9474	10
Tiger Lily	0.9000	0.9000	0.9000	10
Moon Orchid	0.7000	0.7000	0.7000	10
Bird of Paradise	1.0000	0.9000	0.9474	10
Monkshood	0.7692	1.0000	0.8696	10
Globe Thistle	0.8182	0.9000	0.8571	10
... (other classes)				
Total Accuracy	Precision	Recall	F1-Score	Support
Macro Average	0.8499	0.8363	0.8325	1020
Weighted Average	0.8499	0.8363	0.8325	1020

3 Latent Stable Diffusion

This is a Latent Stable Diffusion model consisting of two main components: a Variational Autoencoder (VAE) and a UNet Denoiser. The VAE can encode images into a latent representation and reconstruct them back to images. The UNet denoiser operates in the latent space and learns to step by step refine noisy latent representations. To generate an image from random noise, the UNet denoiser first transforms the noisy latent into a cleaner latent, which is then passed to the VAE decoder to reconstruct the output images.

3.1 VAE Architecture and Training

Although the original task required a standard autoencoder, a Variational Autoencoder (VAE) was implemented for higher-quality image generation. Trained for 60 epochs on fine grained dataset with learning rate of 0.0001, the model achieved per-pixel training and validation losses of 0.0252 and 0.0262 respectively. As shown in Table 3, the VAE has 6,300,547 parameters, with an encoder compressing inputs into a 512-dimensional latent space, and a decoder reconstructing images from it.

Table 3: VAE Architecture (Batch Size=16, Image Size=96×96)

Layer Type	Output Shape	Parameters
Encoder		
Conv2d	[16, 64, 48, 48]	3,136
Conv2d	[16, 128, 24, 24]	131,200
Conv2d	[16, 256, 12, 12]	524,544
Conv2d	[16, 512, 6, 6]	2,097,664
Latent Representation		
Conv2d (mu)	[16, 512, 6, 6]	262,656
Conv2d (logvar)	[16, 512, 6, 6]	262,656
Decoder		
ConvTranspose2d	[16, 512, 6, 6]	262,656
ConvTranspose2d	[16, 256, 12, 12]	2,097,408
ConvTranspose2d	[16, 128, 24, 24]	524,416
ConvTranspose2d	[16, 64, 48, 48]	131,136
ConvTranspose2d	[16, 64, 96, 96]	3075

3.2 VAE Test Performance

The trained VAE achieved a mean per-pixel reconstruction error of 0.006244 with a standard deviation of 0.002823. Figure 2 presents visual examples of test inputs (1st row) and their corresponding reconstructions (2nd row), the output images are blurry but still retain plausible flower outlines, similar colors and spatial layouts, indicating that the model has learned the image patterns within the 512-dimensional latent space.



Figure 2: VAE Test Input and Output Images

3.3 UNet Denoiser Architecture

The UNet denoiser is trained on the latent representations generated by VAE’s encoder using the fine grained dataset. Table 4 shows the model has a total of 27,772,544 parameters, consisting of adaptive noise scaling, three encoder blocks for downsampling, a bottleneck block, three decoder blocks for upsampling the latent representations. Each decoder block incorporates skip connections through channel-wise concatenation with the corresponding encoder block outputs to learn spatial features. Training was conducted 110 epochs in three phases:

Epochs 1–50: learning rate=0.0001, with consistent loss reduction observed

Epochs 51–80: learning rate=0.0001, added time-conditioned positional encoding

Epochs 81–110: learning rate=0.00005, validation timesteps increased from 4 to 10

The trained model achieved training and validation losses of 0.9388 and 0.9418 respectively, suggesting a minor degree of overfitting.

Table 4: UNet Denoiser Architecture (Input Shape = [16, 512, 6, 6])

Layer	Output Shape	Parameters
Noise Scaling		
Linear	[16, 512]	33,408
Encoder Blocks		
Conv2d, GroupNorm	[16, 256, 3, 3]	1,771,080
Conv2d, GroupNorm	[16, 512, 2, 2]	3,542,016
Conv2d, GroupNorm	[16, 512, 1, 1]	4,721,664
Bottleneck		
Conv2d, GroupNorm	[16, 512, 1, 1]	4,721,664
Decoder Blocks		
Conv2d, GroupNorm	[16, 512, 2, 2]	7,080,960
Conv2d, GroupNorm	[16, 256, 4, 4]	2,950,656
Conv2d, GroupNorm	[16, 256, 6, 6]	1,771,008
Channel Adjustment		
Conv2d	[16, 512, 6, 6]	1,180,160

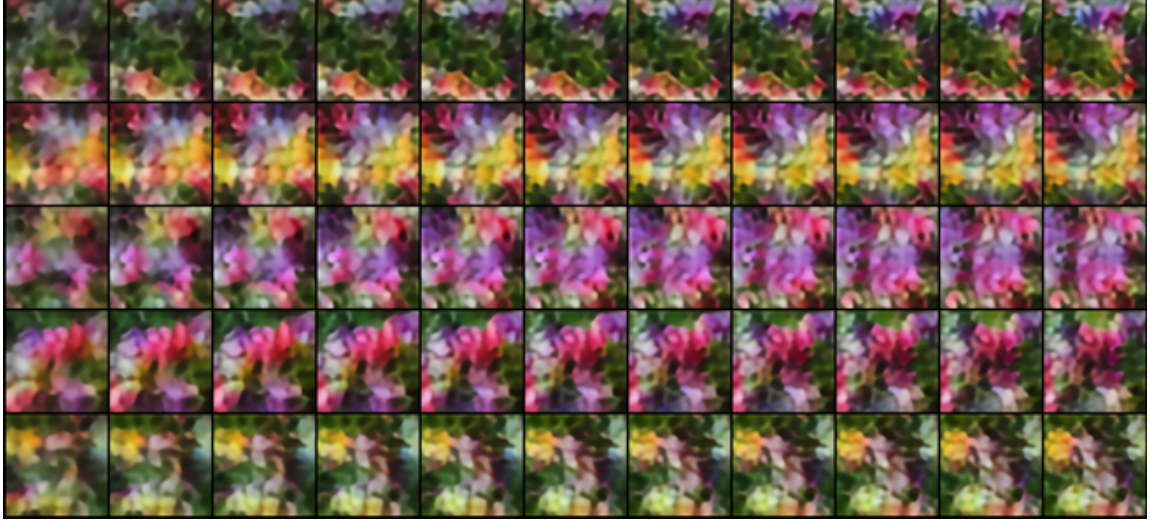


Figure 3: UNet Denoiser Test Generated Images at Various Stages

3.4 UNet Denoiser Test Performance

The trained model achieved an average test loss of 0.9176 over 10 denoising timesteps, suggesting room for further optimization. Figure 3 presents outputs from $t = 1.0$ (full noise) to $t = 0.0$ (cleanest), from left to right showing increasing sharpness and flower-like structures. Even the first timestep resembles colorful blobs, likely because the VAE is well-trained and tends to generate flower-like outputs from random latent noise. However, the current UNet denoiser shows only limited denoising performance, generating high-quality images from noise may require a more enhanced model architecture, a larger dataset, longer training, and more denoising timesteps.

4 Conclusion

To briefly conclude, the CNN classifier, in both coarse and fine grained modes, exhibits a 12–14% training-validation accuracy gap, suggesting moderate overfitting. The test accuracy of 83.63–85.13% indicates reasonable performance but still leaves room for improvement.

For the Latent Stable Diffusion Model, the VAE effectively encodes floral features in the latent space, generating flower-like outputs even from random noise. The UNet Denoiser demonstrates basic denoising capability, with gradual sharpness improvements across timesteps. For future optimization, self-attention and perceptual loss could be explored.